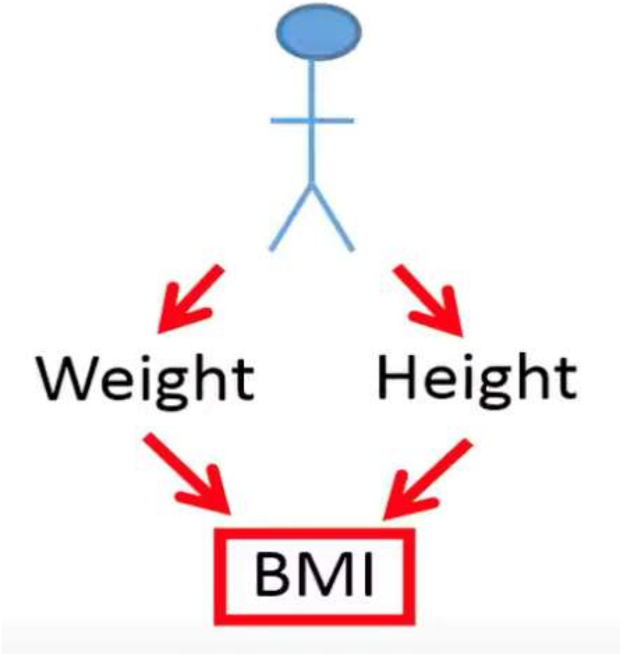
The basics of PCA

- Combining variables
- PCA –basics
- Its applications

Combining variables

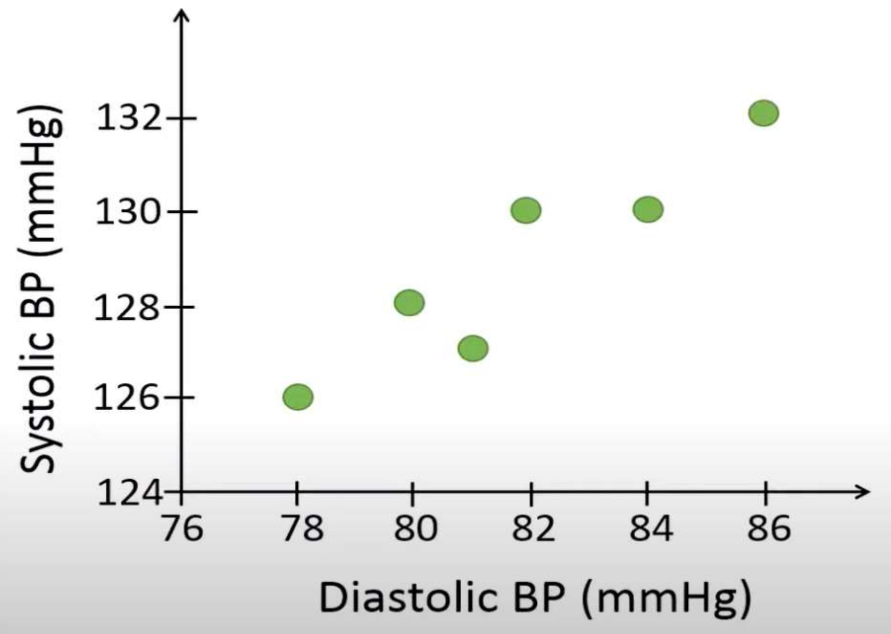


$$BMI = \frac{Weight_{kg}}{Height_m^2}$$

Cholesterol = Weight + Height

Cholesterol = BMI

Combining variables



Diastolic BP	Systolic BP
78	126
80	128
81	127
82	130
84	130
86	132

Combining variables

Diastolic BP	Systolic BP
78	126
80	128
81	127
82	130
84	130
86	132

$$Y = \alpha_1 X_1 + \alpha_2 X_2$$

X1 and X2: two variables to combine
Y: the combined variable
$\alpha 1, \alpha 2$: weights (loadings)

BP = 0.8 DBP + 0.6 SBP

Combining variables

Diastolic BP	Systolic BP
78	126
80	128
81	127
82	130
84	130
86	132

$$BP = \alpha_1 DBP + \alpha_2 SBP$$

$$BP_1 = \frac{DBP_1 + SBP_1}{2} = \frac{78 + 126}{2}$$

$$= 102$$

Combining variables

Diastolic BP	Systolic BP	Mean BP
78	126	102
80	128	104
81	127	104
82	130	106
84	130	107
86	132	109

$$BP = \alpha_1 DBP + \alpha_2 SBP$$

$$BP_1 = \frac{DBP_1 + SBP_1}{2}$$

$$BP_1 = \frac{1}{2} DBP_1 + \frac{1}{2} SBP_1$$

Combining variables

Diastolic BP	Systolic BP	Mean BP	Sum BP
78	126	102	204
80	128	104	208
81	127	104	208
82	130	106	212
84	130	107	214
86	132	109	218

$$BP = \alpha_1 DBP + \alpha_2 SBP$$

$$BP = 1 \cdot DBP + 1 \cdot SBP$$

Conclusion: The difference between two methods is the values used for the weights.

PCA

- Principal component analysis (PCA) is a method to find the linear combination that accounts for as much variability as possible

$$BP = \alpha_1 DBP + \alpha_2 SBP$$

Constraint:

$$\alpha_1^2 + \alpha_2^2 + \ldots + \alpha_p^2 = 1$$

PCA

$$BP = \alpha_1 DBP + \alpha_2 SBP$$

Constraint:

$$\alpha_1^2 + \alpha_2^2 = 1$$

PCA

Diastolic BP	Systolic BP	BP
78	126	138
80	128	140.8
81	127	141
82	130	143.6
84	130	145.2
86	132	148
	Mean =	**142.8**

BP = 0.8 DBP + 0.6 SBP

$$\mathrm{var}(Y) = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{Y})^2 = \frac{1}{5}\left((138-142.8)^2 + \ldots + (148-142.8)^2\right) = \boxed{12.74}$$

PCA

Alpha1	Alpha2	var(Y)
0.8	0.6	12.74
0.6	0.8	11.8
0.98	0.2	10.4
0.2	0.98	7.4

$$BP = \alpha_1 DBP + \alpha_2 SBP$$

Conclusion: We would select the weights 0.8 and 0.6 when we combine the diastolic and systolic blood pressures because these weights generate maximal variance of the combined variables.

PCA

Diastolic BP	Systolic BP
78	126
80	128
81	127
82	130
84	130
86	132

	DBP	SBP
DBP	8.17	5.97
SBP	5.97	4.97

$$Eig = \begin{bmatrix} -0.8 \\ -0.6 \end{bmatrix}$$

$$BP = \boxed{-0.8}DBP + (\boxed{-0.6}SBP)$$

PCA

Example 1

Diastolic BP	Systolic BP	Weight	Height
78	126	67	170
80	128	77	177
81	127	89	183
82	130	90	187
84	130	50	165
86	132	55	164
BP		**BS**	

Cholesterol = BP + BS

PCA

Example 2

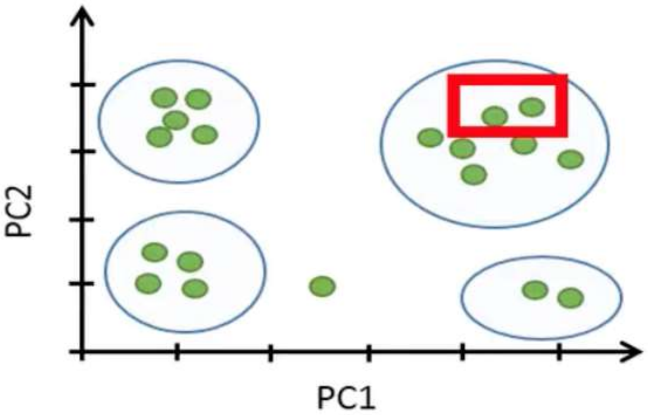
Person	DBP	SBP	BMI	Chol.	Pulse	Temp	...
1	78	126	25	170	55	37.4	...
2	80	128	27	177	56	37.8	...
3	81	127	23	183	60	36.8	...
4	82	130	30	187	61	36.4	...
5	84	130	28	165	62	36.9	...
6	86	132	35	164	70	37	...
...	...	...	...	...	...	...	...

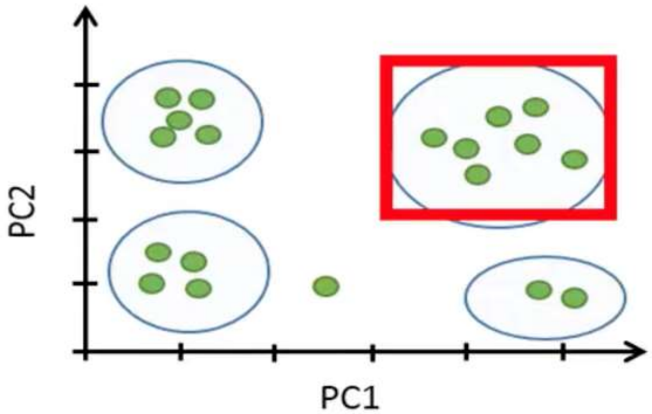
Combine into 2 variables

Person	PC1	PC2
1	3.4	4.5
2	3.7	4.4
3	-21.2	-15.2
4	-20.2	-16.5
5	-8.2	-8.9
6	-0.2	10.2
...	...	...

PCA

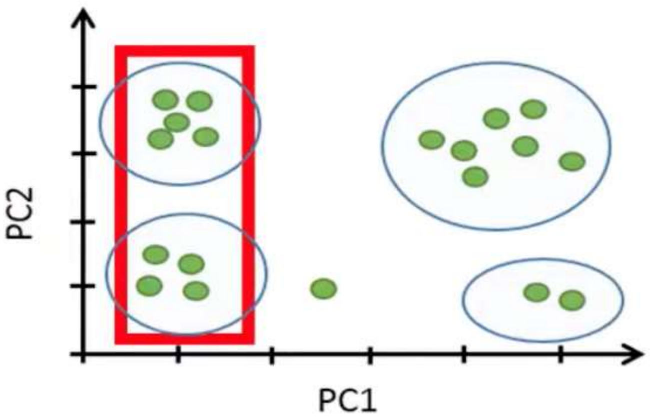
Example 2



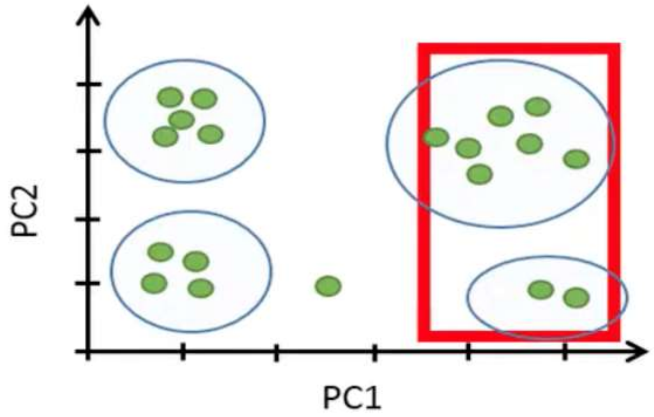


PCA

Example 2



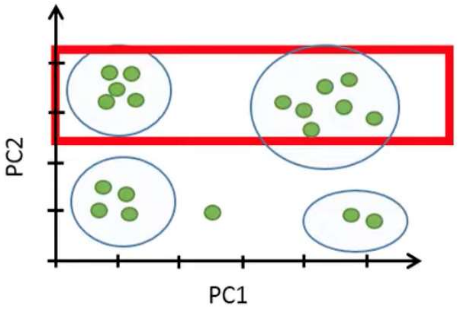
Have similar health profile since these individuals have the same values of the variables associated with the health.



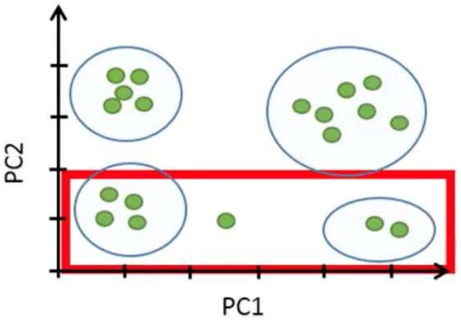
These individuals have a different health profile.

PCA

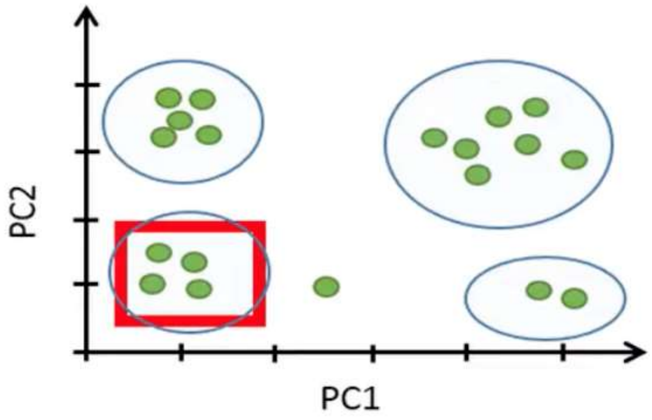
Example 2



These points represent men



These points represent women

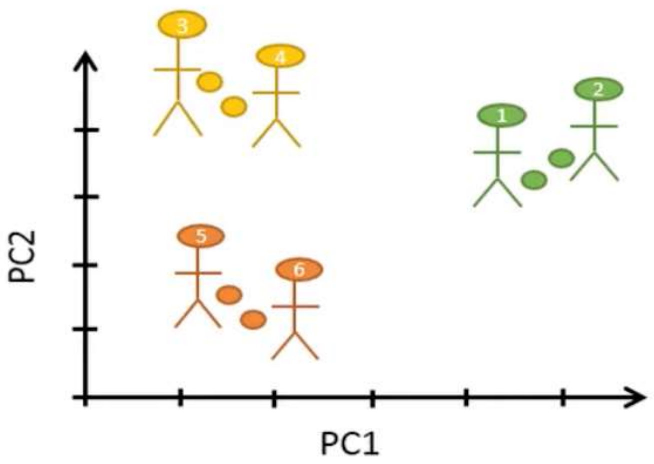


These points represent four women have similar health profile

PCA

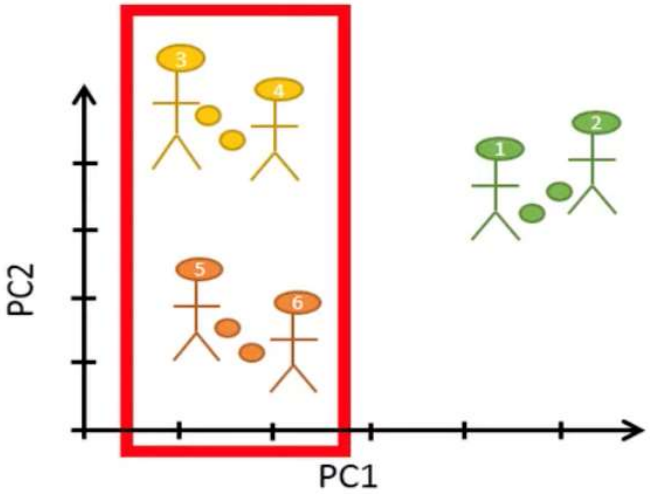
Example 3

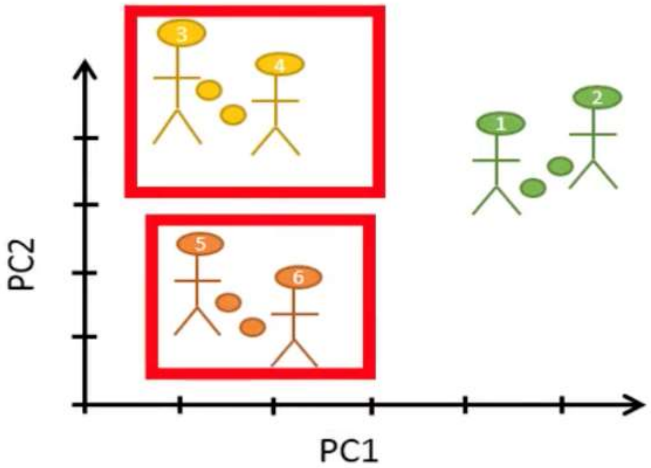
Person	PC1	PC2
1	8	2
2	9	3
3	-5	11
4	-4	10
5	-5	-12
6	-3	-14



PCA

Example 3





PCA

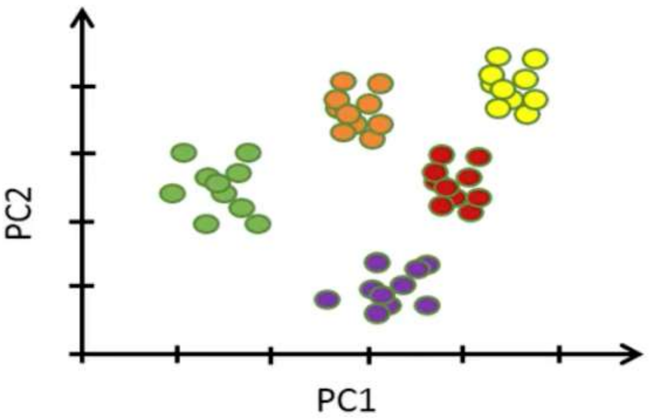
Example 4

Cell	Gene 1	Gene 2	Gene 3	Gene 4	…
Cell 1	0	20	5	0	…
Cell 2	10	0	56	0	…
Cell 3	15	0	20	0	…
Cell 4	10	13	13	0	…
Cell 5	42	55	60	0	…
Cell 6	0	0	30	0	…
…	…	…	…	…	…

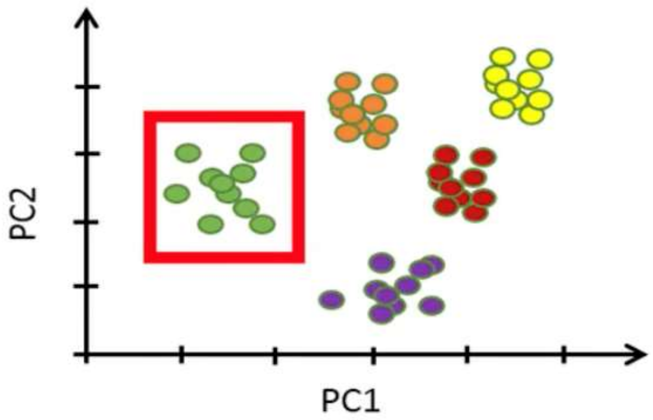
Cell	PC1	PC2
Cell 1	5	22
Cell 2	-10	0
Cell 3	-3	13
Cell 4	-3	13
Cell 5	19	55
Cell 6	-7	-7
…	…	…

PCA

Example 4



Having 50 points represent 50 cells



Cells can be found in a distinct cluster

Reference

- Tilevik, A. (n.d.) PCA1: Basics – simply explained. Available online: https://www.youtube.com/watch?v=dz8imS1vwIM&t=3s